Explaining the better move

28 rounds of hill-climbing a game review's lessons, graded on the board and by a blind judge

Chess Coach research lab

Abstract

At each key moment of a game review, the coaching app explains what the move played allowed, what the engine's better move does instead, and the difference between them. We measured those explanations on the 329 key moments of 60 of the player's games, and the later rounds also on endgame sets of 160 and 130 more games, with three grades checked against the board, and changed the code that writes them one idea per round, keeping a round only when no grade fell on either half of the games (one round fixed five text defects at once). Where a change wrote something the grades cannot check, a judge model compared the old and new texts blind, twice, with the order reversed. Over 28 rounds (25 kept), the share of held-out moments whose texts make no false checkable claim rose from 90.1% to 100.0%, and the share that names a concrete consequence of the better move from 75.0% to 82.9%. The judge preferred the texts of every kept round it read in most of the comparisons it decided on each half, and a length control showed that was not for their length: filler as long as a round's change won 0 of 234 comparisons against the same text without it. Each kept round is a change to the app's own code. The lab recomputes every published number from the records, and the app rerun on fresh engine evidence, carrying the kept rounds through round 29, scores 100.0% and 85.0% on the same two measures.

1 What the player reads

When a game review stops at a key moment (a mistake, a blunder, a missed win), the app shows a card with a headline and a verdict, a lesson panel (what the move played allowed, what the better move does instead, the difference between the two lines, a takeaway), a Why sheet with the evidence behind the verdict, and a retry: the player may try the position again, with a hint and then the answer. Every one of those texts is written by heuristics in the app's review and knowledge services from the engine's evidence. No language model writes them, so every claim they make can be traced to a template, and most can be checked on the board.

The player is an adult club player, rated 948–1167 on chess.com across the 60 games studied here. Two things matter: that the texts are true, and that they say *why* the better move is better, in terms the player can use at the board, not only by how much.

2 Data and measures

Moments. The 329 key moments the review stops at in 60 of the player's games, from the app's phone-profile engine evidence for each game. The games are split 30 for train and 30 for test, the same split as the accuracy project, so no game is train in one and test in the other: 177 train moments and 152 test moments. A round was designed on train moments only.

Grades. Each moment's texts get three grades, all checked against the board with chess.js and against the stored engine lines (`grade.mjs`):

- **faithful**: 1 unless a checkable claim in the visible text is false: an illegal move, a line that does not replay, a piece said to attack what it does not, material said to be won that the line does not win, a mate that is not there, or an "if they take it with" answer that is not the engine's line or whose claim does not hold at its end.
- **concrete**: 1 when the better move's explanation names a consequence (material, a mate, a named threat, a prevented reply) and the text says what the move played allowed or missed.
- **named_deep**: 1 when every move the text names as best rests on a search at the app's best budget. It is a property of the evidence, not of the words, and is held, not targeted.

A fourth grade, **faithfulAll**, is faithful over the lesson the Why sheet also shows on a moment whose card shows only a comment. A clause no template recognises is reported as unparsed, never graded true. The grader was checked by hand on 19 moments before any round; the hand verdicts and the baseline's grades agree on all three grades for 19 of them.

Every rate below is a share of key moments, with a 95% interval from a bootstrap over games (a game's moments share its evidence, so the game is the unit). A variant's difference from the baseline is paired on the same moments.

The judge. The grades cannot see whether a new clause *helps*. For a round that writes something new, every moment whose texts change is shown to a judge model as two versions (X and Y, by a seeded coin), each laid out as the app lays it out, with the position, the move, the verdict, the engine's evaluation and best line. The judge answers which version would help the player more (understand why the better move is better, and know what to look for next time), or that they are the same, with one sentence of reason. It answers twice, with X and Y swapped the second time. The rule, fixed before any run: keep the round if, pooling both runs, the new texts are preferred in more than half of the decided comparisons on train *and* on test.

3 The rounds

Table 1 lists every variant. Rounds 1 and 2 fix false claims the baseline's templates made, found by the grader over all moments before the split was used to design anything. Rounds 3 to 10 add something the player asked for on the app's Coach Calibration page or that the judge's notes named, so the judge decides them. Round 11 fixes five defects in round 10's texts that the player's calibration answers and the judge's notes flagged; each can be read off the text and the board, so the round counts them rather than asking the judge. Round 12 answers the question the player's second set of calibration answers kept asking, what happens when the opponent simply saves the attacked piece, so the judge decides it. Round 13 shortens round 12's sentence where the judge's notes said it ran long, so the judge decides it too. Round 14 corrects how the lessons name an amount: its number was the line's net, but the pieces it named were not always the ones that changed hands. That is read off the board, which the judge does not see, so the round counts its defects as round 11 did. Round 15 fills the sections on what the played move allowed that named the reply and said nothing about it, which the player's own calibration answers had pointed to, so the judge decides it. Round 16 tells an endgame's passed pawns' story, which the player's calibration answers asked for in his rook endings; the 60 games hold too few endgame moments to decide it, so it is also graded and judged on an endgame set of 160 more games (Section 6), and the judge decides it. Round 17

says where each move leaves the game, which the player's third set of calibration answers asked about; it changes 347 moments, more than a judge can read, so the judge reads a seeded sample of them and decides it. Round 17c makes the card's folded note on how its grade was made give each move the number round 17's sentence gives it; the judge does not see that note, so the round counts its defect as rounds 11 and 14 did. Round 18 says what a better check does, which the player's answers to the coach's questions on what a lesson should explain asked about; the first two sets change few moments with it, so it is also graded and judged on a second endgame set, and the judge decides it. Rounds 19 and 20 come from the same answers: round 19 says the guard a played move gives up, and round 20 says when the piece a move attacks escapes to a better square. Each changes more moments than a judge can read, so the judge reads a seeded sample of them, as in round 17, and decides it. Round 20's first version failed its grades before any judge ran, and its variant 20b is the round judged. Round 21 comes from the same answers: it says what decides the pawn ending a trade leads to. It changes few moments, so the judge reads every one and decides it. Round 22 comes from them too: it says what the better line does first when the order of its moves matters. It changes few moments, so the judge reads every one. Its first wording joined two sentences with a semicolon, which the app's rules for its words forbid, so its variant 22b says the same words as two sentences and was judged again, and round 22 is decided on both. Round 23 comes from the same answers, by way of the lab's study of the move after an opponent's error (`projects/book-advice`), which finds traps a real share of the player's failures to punish: it says when a capture that looked free was a trap. It changes few moments, so the judge reads every one and decides it. Round 24 comes from the answers again: when a pawn queens in one line and not the other, it says where the other side's passed pawn stands in that race. It changes few moments, so the judge reads every one and decides it. Round 25 comes from the answers too: in an endgame, it names the square the better move keeps their king out of. It changes few moments, so the judge reads every one and decides it. Round 26 comes from a count with its bar written first: of the 414 training lessons whose better move is quiet, 87 (21.0%) gave no reason for it, past the bar (69 of 376, 18.4%, on the test halves, counted after the bar decided). The answers say that where no material or mate explains the engine's move, the review should look at activity, restriction and longer-term pressure; round 26 reads the first, the piece the better move brings into play. It changes few moments, so the judge reads every one and decides it. Rounds 28 and 29 come from the player's question on a lesson that preferred a king's step to blocking a check, whose difference said the engine's reason was not established. Round 28 corrects what reading the lessons around it found: the same sentence stood where a later sentence gave the reason. That is a mistake to correct, not an idea to test, so the round counts it as rounds 11, 14 and 17c did. Round 29 gives his lesson its reason, a pin that round 19's guard sentence did not see. It changes few moments, so the judge reads every one and decides it. No round 27 ran: its count, with its bar written first, found too few of the player's moves to judge.

Round 1: each line's material as it is.

The lesson's fallback sentence said "neither line wins material in the next 5 moves" whenever no earlier branch fired, including when both lines won the same material. It now says each line's material.

Round 2: the seventh-rank rook.

The insight named pawns the rook could not see past another pawn. It now names only the pawns the rook hits.

Round 3: what the better move defends.

A piece the reply takes, guarded or moved away by the better move (Figure 1).

Round 4: the capture a learner asks about first.

Where the better move or the reply lands on a square it can be taken on and the engine's line

		test (%)		train (%)		texts	judge
variant	change	faithful	concrete	faithful	concrete	changed	train / test
baseline	the app before the rounds	90.1	75.0	87.6	78.0		
v1	Say each line's material as it is in the lesson's difference	96.1	82.2	98.3	84.7	20, 34	
v2	Name only the pawns the seventh-rank rook hits	100.0	82.2	99.4	84.7	25, 35	
v3	Say what the better move defends	100.0	82.2	99.4	84.7	31, 43	22–0 / 18–2
v4	Answer the capture a learner asks about first	100.0	82.9	99.4	84.7	34, 45	8–0 / 6–0
v5	Say where a long mate ends (*not kept*)	100.0	82.9	99.4	84.7	45, 54	6–6 / 14–2
v6	Say where a long mate ends, when the end is in reach	100.0	82.9	99.4	84.7	44, 51	4–0 / 10–0
v7	Say what a capture the better line declines would cost (*replaced by v7b*)	98.7	82.9	97.2	84.7	65, 77	48–3 / 40–1
v7b	Say what a capture wins, not the balance it leaves	100.0	82.9	99.4	84.7	65, 77	51–0 / 41–0
v8	Say when the played move is right but too early	100.0	82.9	99.4	84.7	70, 87	50–0 / 20–0
v9	Answer the natural capture, and skip what a pawn plainly guards (*not kept*)	100.0	82.2	98.9	85.9	142, 166	1–35 / 0–12
v10	Say the natural capture that only trades as the trade it is	*graded on rebuilt documents (Table 4)*					18–2 / 12–0
v11	Five text defects, fixed where the texts are written	*graded on rebuilt documents (Table 4)*					
v12	Say what the natural defense to the better move costs	*graded on rebuilt documents (Table 4)*					19–9 / 22–12
v13	Leave out the moves the defense sentence does not need (*not kept*)	*graded on rebuilt documents (Table 4)*					0–16 / 4–24

Table 1: Every variant. Rates are shares of key moments; *texts changed* counts the test and train moments whose visible texts differ from the baseline's; *judge* is the round's pooled preferences, new–old, on train and test (Table 2). The later variants follow.

		test (%)		train (%)		texts	judge
variant	change	faithful	concrete	faithful	concrete	changed	train / test
v14	Say an amount in the pieces it is made of	*graded on rebuilt documents (Table 4)*					
v15	Say what a quiet reply defends	*graded on rebuilt documents (Table 4)*					22–1 / 8–0
v16	Tell an endgame's passed pawns' story	*graded on rebuilt documents (Table 4)*					37–2 / 18–3
v17c	Say where each move leaves the game	*graded on rebuilt documents (Table 4)*					73–0 / 76–0
v18	A better check is said by the guard it takes off a square	*graded on rebuilt documents (Table 4)*					20–0 / 13–1
v19	The guard a played move gives up	*graded on rebuilt documents (Table 4)*					96–1 / 90–0
v20b	The chased piece that escapes to a better square	*graded on rebuilt documents (Table 4)*					87–3 / 84–2
v21	The pawn ending a trade leads to	*graded on rebuilt documents (Table 4)*					10–0 / 11–1
v22b	What the better line does first	*graded on rebuilt documents (Table 4)*					20–0 / 26–0
v23	A capture that was a trap	*graded on rebuilt documents (Table 4)*					28–6 / 29–3
v24	Where the other side's pawn is when one queens	*graded on rebuilt documents (Table 4)*					8–6 / 6–2
v25	The square the better move keeps their king out of	*graded on rebuilt documents (Table 4)*					14–0 / 6–0
v26	The piece the better move brings into play	*graded on rebuilt documents (Table 4)*					22–0 / 18–0
v28	No absence said before a reason	*graded on rebuilt documents (Table 4)*					
v29	The pin that keeps a piece from taking the reply	*graded on rebuilt documents (Table 4)*					12–0 / 4–0

Table 1: Every variant, continued.

takes it, the lesson says what follows (Figure 2).

Rounds 5 and 6: where a long mate ends.

A mate is told with where it ends and what holds it there. Round 5 did so for every mate of three moves or more and failed its rule on train; round 6 keeps it for mates of three to seven moves with no promotion on the way.

Round 7: a capture the better line declines.

A search of each capture the line declines (`probes.mjs`, at the routine's budget), and the lesson says what it would cost: a recapture as the trade it is, or checks that win a piece, move by move (Figure 3). Round 7b fixes the material clause the judge found false in a few moments (a gain read as a balance).

Round 8: the right move, too early.

Where the move played is in the engine's best line at the mover's next or second turn, the difference and the takeaway say so, with the line (Figure 4).

Round 9 (not kept): fewer capture answers.

Skipping an answer when a pawn plainly guards the square, and stopping at the first capture with no answer. The judge wanted every answer back (Figure 5).

Round 10: the natural capture that only trades.

The cheapest capture of the better move's piece is said as the trade it is when its search takes back at once on the same square, nothing takes there again, and the material ends where it began: "if they take it with axb5, axb5 takes back: an even trade" (Figure 6). The round was graded on documents rebuilt for it (Section 6).

Round 11: five text defects.

Five errors in round 10's texts, fixed together because each is a mistake to correct, not an idea to test: the retry's answer opened its later sentences in lower case; a reply that only takes back what the move captured was called a win "outright"; a capture or a check was called "a quiet improvement"; the difference called the lines "too short or unsettled" to compare beside a capture the same card calls outright; and a line's net gain was glossed with a piece worth more than it, as in "they win a piece and a pawn (your rook on d7)". The round was graded on round 10's rebuilt documents (Section 6).

Round 12: what the natural defense costs.

When the better move threatens a piece, a human-move model gives the defender's likeliest reply at the defender's rating band. When that reply moves the attacked piece or blocks the attack, the engine's line from it (the better line itself when it is the engine's reply there, otherwise a search at the routine's budget, `defenses.mjs`) says what it costs: "if they move the rook (Rf7), Qxf5, and you have won a pawn" (Figure 7). A capture of the attacker is left to the capture answers, and the sentence appears only when the line wins material after the defense or mates. The round was graded on documents rebuilt for it (Section 6).

Round 13 (not kept): the defense sentence, shorter.

Where the defense is the engine's own reply, the sentence points to the engine's line instead of printing its moves again ("if they move the bishop (Bb2), as in the engine's line, you have won a pawn by move 14"); where the line would show more than 4 moves, it says only when the gain is in ("if they move the queen (Qxd5), you have won about 6 pawns of material by move 19"). The judge wanted the moves back (Figure 8).

Round 14: an amount in the pieces it is made of.

The lesson turned a line's net material into a count whatever pieces made it up, so a bishop won for two pawns was "a pawn" and a rook won for a pawn "a piece and a pawn"; the same count picked the piece said to fall, and some lines gave that piece back. Each line now carries the pieces its mover won and gave, by kind, with trades set off (a piece against one of its kind,

then a knight against a bishop), and an amount keeps its count only when the count names those pieces. Otherwise it names them: "they win a bishop for two pawns", "limits the loss to a rook for a pawn". Where one sentence gives an amount for two lines whose pieces differ, it says the worth ("a pawn's worth of material"), and a piece is said to fall only when the line keeps it. The round was graded on round 13's documents (Section 6).

Round 15: what a quiet reply defends.

Where the section on what the played move allowed named the reply and gave no reason for it, and the reply captures nothing, the lesson says what the reply defends, checked on the board: it moves or guards the piece the better move would have won or attacked ("it moves the rook off e5, where dxe5 would have won it"; "it guards the pawn on e4, which Nxe4 would have won"), takes away the better move's check ("it takes away the check Qd3 would have given"), or answers the played move's own threat ("it guards the pawn on b5, which Rd5 attacked"). The headline's sentence about the reply carries the point. The round was graded on the documents round 14 was (Section 6).

Round 16: an endgame's passed pawns.

In an endgame, and when neither line ends in mate, the lesson follows each pawn over the lines it reads. The pawn that queens after the played move and not after the better one opens the section on the difference with its steps and its fate in the other line ("After Kg4, their b-pawn queens: b7 and b8=Q in the engine's line. Nxb6 takes it first."; "After Rc3 it falls (Rxh2)"); a reply that pushes that pawn "goes on to queen"; a better move that takes a passed pawn, or stands in the path of one past the middle of the board, says so ("it blocks their passed b-pawn on b7"); and a king's walk to a passed pawn, each move closer, is named with its moves ("your king walks over to their c-pawn: Kb2, Kc3 and Kxc4"). The shown lines run on to the promotion or the walk's last move. The round was graded on the documents round 15 was and on the endgame set (Section 6).

Round 17: where each move leaves the game.

On an inaccuracy, mistake, miss or blunder, the section on the difference ends with where the played move and the better one leave the mover, in the words and the number the app gives a score, from the mover's side. When both moves leave the mover ahead (much better or winning) or behind (much worse or losing), it says so ("You are behind either way: losing after Rb5 (−8.1), much worse after Rxb2+ (−4.4)."); on a mistake, miss or blunder whose better move leaves a better standing, it says where each leaves the mover ("After Kg4 you are losing (−7.0). After Nxb6 you would be winning (+7.0)."). It says nothing when both moves leave the game open, from worse to better. Round 17 first wrote the second form as one sentence joined by a semicolon, which the app's rules for its words forbid; round 17b writes it as two. The round was graded on the documents round 16 was, in both sets (Section 6). The card's folded note on how its grade was made gives the same two scores, and it rounded them to a tenth its own way, so the note and the new sentence could give a move numbers a tenth apart; round 17c gives the note the sentence's numbers.

Round 18: a check with a reason.

When the better move gives check, the king's reply in its line steps off a square the king guarded, and the mover's next move lands there, the section on the difference says so: "Rd8+ is a check with a reason: after Kh7, their king no longer guards f8, and Rf8 follows." ("and Rxf1+ takes the rook there" when that move captures). The round was graded on the documents round 17 was, on every key moment of the endgame set, and on a second endgame set (Section 6).

Round 19: the guard a played move gives up.

When the piece the played move moved guarded the square the engine's reply lands on, nothing

of the mover's guards it after the move, and the guard held something, the section on the difference says so: "After Qe1, your queen no longer guards d3, and Rxd3 takes the bishop there." ("After Kc1, your king no longer guards c3, and Kc3 follows." when the reply takes nothing). The round was graded on round 18's three sets (Section 6).

Round 20: the chased piece that escapes.

When the played move gives no check and newly attacks a piece with a threat, and the engine's reply moves that piece away to take something, give check or attack something new, the section on the difference says so: "Ne5 attacks their bishop on g4, but it escapes to f5, where it attacks your queen on d3." ("Nf3 attacks their queen on h4, but it escapes by taking your bishop on f4." when it escapes by taking). A reply that takes the attacker, a trade and the better line's own answer do not count. The first version took any attacker worth less than the piece as a threat, and the app counts a king as worth nothing, so it called a king's attack on a guarded piece a threat; round 20b counts a king's attack only on an unguarded piece. The round was graded on round 19's three sets (Section 6).

Round 21: the pawn ending a trade leads to.

When the better move's line or the played move's leaves only kings and pawns within the plies a lesson reads, from a position with another piece, the section on the difference plays the ending on and says what each king takes and which pawn queens first: "After Qxg1+ Kxg1, only kings and pawns are left: in the engine's line your king takes their pawns on h4 and g2, and their king takes your pawn on e3." ("After Qxh2+ Kxh2, only the kings are left: a draw." when no pawn is left). The round was graded on round 20's three sets (Section 6).

Round 22: what the better line does first.

When the better line plays the played move one or two of the mover's moves later and the engine's reply to the played move captures, the section on the difference says which capture the played move, made now, allows, and what the line does first to stop it: it takes the capturing piece ("Played now, Bf5 allows dxc5. cxd4 first takes the pawn on d4."), drives it off with an attack ("Played now, Nc6 allows Qxb7. Qd7 first drives the queen from b5 (Qe2)."), or moves the target away ("Played now, g6 allows Bxf5. Nfd6 first moves your knight on f5 away."). It says so only when the target is not taken on the way and nothing can take it once the played move comes. Round 22 first joined the two sentences with a semicolon, which the app's rules for its words forbid; round 22b writes them as two. The round was graded on round 21's three sets (Section 6).

Round 23: the capture that was a trap.

On a mistake, blunder or miss that takes a piece nothing can take back, not en passant, when the better move lands elsewhere and the engine's line after the capture mates the mover, or ends with nothing pending two pawns' worth or more down while the better line ends two pawns' worth or more better, the headline opens with the trap: "Their pawn on c7 looked free, but it was a trap. Qxc7 allows Rxe6, and it wins the rook outright." ("Their rook on h7 looked free, but it was a trap. Rxh7 allows mate in 1 with Bf7#. Ng4+ avoids it." before a mate). The round was graded on round 22's three sets (Section 6).

Round 24: where the other side's pawn is when one queens.

When the difference tells a pawn that queens in one line and not the other, and at that promotion the other side has a passed pawn, the pawn's story ends with where the one nearest its last rank stands, how many squares it has left and the first piece on its path: "When h1=R comes, your b-pawn is still on b5, three squares from b8, with your knight on b6 in its way." ("When a1=Q comes, your h-pawn is still on h6, two squares from h8." when its path is clear). The round was graded on round 23's three sets (Section 6).

Round 25: the square the better move keeps their king out of.

In an endgame, when the engine's reply to the played move is their king's step to a square nothing of the mover's covered, and the better move covers that square with the piece it moves while their king never stands there in the better line, the better move's points say so: "it keeps their king out of c5". It comes after the board's readings that rank above it, such as the opposition, and before the rest, so as the first point it gives the headline's reason, and the headline opens "Kd6 was better: it keeps their king out of c5. After Kf6, they can answer Kc5". The round was graded on round 24's three sets (Section 6).

Round 26: the piece the better move brings into play.

When nothing else explains the better move of an inaccuracy, mistake, blunder or miss (its section has no point, its line ends with no more material than the played one, and the headline does not name the reply's threat), and the better move takes nothing from a position where the mover is not in check, the better move's points say which of the mover's knights, bishops, rooks and queens it brings into play: one that had at most three squares to go to (its legal moves) before the move, has at least three more after it, and at least three more than the played move leaves it; the largest gain, and the piece the move moves on a tie: "it brings the bishop on c8 into play, with 5 squares to go to instead of none". As the section's only point it gives the reason in the headline, which reads "d5 was better: it brings the bishop on c8 into play, with 5 squares to go to instead of none.", in the difference and in a takeaway such a lesson did not have. The round was graded on round 25's three sets (Section 6).

Round 28: no absence said before a reason.

When the difference opens with the lines' material and names nothing about the better move, that sentence ends "The concrete reason for the engine's preference is not established." The difference's later sentences can give a reason all the same (the guard the played move gives up, the chased piece's escape, the check with a reason, why the order matters, the pawn that queens, the pawn ending a line reaches), so the card said it had found no reason and then gave one. Where one of them is written, the absence is left out: "The lines shown do not establish a material advantage for either continuation. After c5, your pawn no longer guards b5, and Bb5+ follows." The round was graded on round 26's three sets (Section 6).

Round 29: the pin that keeps a piece from taking the reply.

Round 19's guard sentence needs nothing of the mover's to attack the reply's square after the move, and a guard the played move puts in a pin still attacks it. The sentence now names the pin when the moved piece is the one piece of the mover's that attacks the square from where it went, an enemy piece pins it there to the king (it was not pinned where it stood), and after the reply it has no legal move to the square: "After Be7, your bishop no longer guards h4: the rook on e1 pins it to your king, and Nh4 follows." And the section on what the played move allowed gains a point, after its others, when a piece of the mover's that attacks the reply's square cannot take there because an enemy piece pins it to the king, the capture would win material, the king is in check from nothing but the reply, and no legal capture there wins material: "it is safe from your knight on d2: the bishop on b4 pins the knight to your king". The round was graded on round 28's three sets (Section 6).

round	judged	train	test	the rule says
3	round 3	22–0 (0)	18–2 (0)	keep
4	round 4	8–0 (0)	6–0 (0)	keep
5	round 5	6–6 (10)	14–2 (12)	do not keep
6	round 5 (17 moments)	4–0 (8)	10–0 (12)	keep
7	round 7	48–3 (15)	40–1 (9)	keep
7b	round 7b	20–0 (0)	12–0 (0)	keep
7 and 7b	round 7 then round 7b	51–0 (15)	41–0 (9)	keep
8	round 8	50–0 (0)	20–0 (0)	keep
9	round 9	1–35 (0)	0–12 (0)	do not keep
10	round 10	18–2 (0)	12–0 (0)	keep
12	round 12	19–9 (0)	22–12 (0)	keep
13	round 13	0–16 (0)	4–24 (0)	do not keep
15	round 15	22–1 (1)	8–0 (0)	keep
16	round 16	37–2 (3)	18–3 (1)	keep
17	round 17 (80 moments)	72–0 (8)	74–0 (6)	keep
17b	round 17b (49 moments)	52–0 (0)	46–0 (0)	keep
17 and 17b	round 17 then round 17b (80 moments)	73–0 (7)	76–0 (4)	keep
18	round 18	20–0 (0)	13–1 (0)	keep
19	round 19 (94 moments)	96–1 (1)	90–0 (0)	keep
20b	round 20b (88 moments)	87–3 (0)	84–2 (0)	keep
21	round 21 (11 moments)	10–0 (0)	11–1 (0)	keep
22	round 22 (23 moments)	20–0 (0)	24–0 (2)	keep
22b	round 22b (23 moments)	20–0 (0)	26–0 (0)	keep
22 and 22b	round 22 then round 22b (23 moments)	20–0 (0)	26–0 (0)	keep
23	round 23 (33 moments)	28–6 (0)	29–3 (0)	keep
24	round 24 (11 moments)	8–6 (0)	6–2 (0)	keep
25	round 25 (10 moments)	14–0 (0)	6–0 (0)	keep
26	round 26 (20 moments)	22–0 (0)	18–0 (0)	keep
29	round 29 (8 moments)	12–0 (0)	4–0 (0)	keep

Table 2: The judge rounds, pooled over both runs: new–old preferred, with the same-answers in parentheses. Round 6 made no run of its own: it reads round 5's judgments on the moments whose texts round 6 changes. Round 17's judges read a seeded sample of the moments it changed; round 17b judged again the sampled moments whose wording it changed, and round 17 is decided on both. Rounds 19, 20b, 21, 22, 23, 24, 25, 26 and 29 were judged on seeded samples too, round 21's, round 22's, round 23's, round 24's, round 25's, round 26's and round 29's taking every moment they changed; round 22b judged every moment round 22 changed again, on its final wording, and round 22 is decided on both, as round 17 is. Round 20's first version was not judged.

4 Is the judge rewarding length?

Before round 10, the judge picked the longer of its two versions in 305 of the 326 comparisons it decided in rounds 3, 4, 5, 7, 7b, 8 and 9. A judge that simply prefers longer text would have voted the same way, so a length check, fixed before any of its answers, tested that on the moments of rounds 7, 8 and 9. Each moment gave three comparisons: the longer text against itself; a *decoy*, the shorter text with filler about as long as the round's change, against the shorter text; and the longer text against the decoy. The filler either restates what the card already shows (its line, or that the move is the engine's first choice) or is advice that fits any move. The last two comparisons were shown in both orders, and the 585 items were dealt into 20 batches under opaque ids, each read by a fresh judge with the rounds' instructions, so no judge could tell a control from a round.

The judge answered "same" to 117 of 117 identical pairs, and preferred the decoy in 0 of 234 comparisons with the shorter text (Table 3). The rounds that added words won on what the words said. From round 10 on, each round's batches also carry a padding control: moments the round does not change, set against themselves with filler as long as the round's change. If the decoy wins at least a quarter of those comparisons and more than half of the decided ones, the judge has drifted toward length and the round is not kept on its word. In round 10 the decoy won 0 of 20, and in round 12 0 of 20. Round 13 set out to shorten its texts, so its batches carry the mirror, a trim control: moments the round does not change, set against themselves with the last point of the better move's section cut. A judge drifting toward short texts would prefer the cut one, under the same rule; it did in 1 of 20 comparisons (16 the full text, 3 same). Round 15 grows the section on what the played move allowed, so its padding control pads that section, on moments where it says nothing and the round leaves it so; the decoy won 0 of 20 comparisons (17 the shorter text, 3 same). Round 16 grows several sections of the card at once, so its padding control adds a sentence of filler after the difference, as long as the round's median growth of the whole card, on endgame moments the round leaves unchanged; the decoy won 0 of 20 comparisons (14 the shorter text, 6 same). Round 17 adds one sentence at the end of the difference, so its padding control does as round 16's did, on moments of the endgame set the round leaves unchanged; the decoy won 0 of 20 comparisons (0 the shorter text, 20 same). Round 18 adds one sentence to the difference, so its padding control does the same; the decoy won 0 of 20 comparisons (0 the shorter text, 20 same). Rounds 19, 20b, 21 and 22 each add one sentence to the difference as well, and their padding controls do as round 18's did: the decoy won 0 of 20 comparisons in round 19 (10 the shorter text, 10 same), 0 of 20 in round 20b (4 the shorter text, 16 same), 0 of 20 in round 21 (20 the shorter text, 0 same) and 0 of 20 in round 22 (0 the shorter text, 20 same). Round 22b adds no length to round 22's texts, so it carries no control of its own. Round 23 adds its sentence before the headline, so its padding control puts the filler there, as long as the round's median growth of the headline; the decoy won 0 of 20 comparisons (20 the shorter text, 0 same). Round 24 adds one sentence to the difference again, and its control does as round 18's did: the decoy won 0 of 20 comparisons (10 the shorter text, 10 same). Round 25's point shortens the card on six of the ten moments it changes, where it replaces a longer reading as the first point's reason or moves one past the section's three, so the card's median growth is below zero and its control's filler among the better move's points is the shortest it has, 32 to 44 characters: the decoy won 0 of 20 comparisons (10 the shorter text, 10 same). Round 26's point also gives the reason in the headline, the difference and the takeaway, and the retry card shows it, so the card grows by a median of 549 characters (508 to 565) over the moments it changes, longer than any filler the control has; its filler among the better move's points stops at the longest it has, 95 to 155 characters: the decoy won 0 of 20 comparisons (20 the shorter text, 0 same). Round 28 was not judged, so it carries no control. Round 29's point stands in the section on what the played

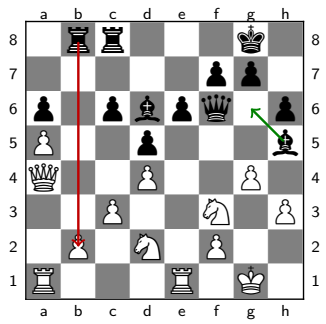
move allowed, and its plan let the point reach the headline, so its padding control pads that section with filler as long as the round's median growth of the whole card, 80 characters (77 to 82 over the moments it changes; the filler runs 79 to 91): the decoy won 0 of 20 comparisons (20 the shorter text, 0 same). Its five restating decoys read as wrong, not only as long: to reach that length their filler adds to the section's line, which opens with the played move, that it is the engine's first choice, and every reason the judges gave on them names those words. Round 15's control, in the same section, was shorter and restated the line alone. On the five neutral decoys alone, the shorter text won 10 of 10 comparisons.

check	result	verdict
J1, the same text on both sides: answered "same"	117 of 117	uses "same"
J2, the shorter text against itself with filler: the decoy preferred	0 of 234 (232 the shorter text, 2 same)	not length-biased
J3, round 7: the longer text against a decoy of its length	train 66 of 66, test 50 of 50	survives
J3, round 8: the longer text against a decoy of its length	train 50 of 50, test 20 of 20	survives
J3, round 9: the longer text against a decoy of its length	train 36 of 36, test 12 of 12	survives
padding control, round 10: the decoy preferred	0 of 20 (20 the shorter text, 0 same)	no drift
padding control, round 12: the decoy preferred	0 of 20 (20 the shorter text, 0 same)	no drift
trim control, round 13: the decoy preferred	1 of 20 (16 the full text, 3 same)	no drift
padding control, round 15: the decoy preferred	0 of 20 (17 the shorter text, 3 same)	no drift
padding control, round 16: the decoy preferred	0 of 20 (14 the shorter text, 6 same)	no drift
padding control, round 17: the decoy preferred	0 of 20 (0 the shorter text, 20 same)	no drift
padding control, round 18: the decoy preferred	0 of 20 (0 the shorter text, 20 same)	no drift
padding control, round 19: the decoy preferred	0 of 20 (10 the shorter text, 10 same)	no drift
padding control, round 20b: the decoy preferred	0 of 20 (4 the shorter text, 16 same)	no drift
padding control, round 21: the decoy preferred	0 of 20 (20 the shorter text, 0 same)	no drift
padding control, round 22: the decoy preferred	0 of 20 (0 the shorter text, 20 same)	no drift
padding control, round 23: the decoy preferred	0 of 20 (20 the shorter text, 0 same)	no drift
padding control, round 24: the decoy preferred	0 of 20 (10 the shorter text, 10 same)	no drift
padding control, round 25: the decoy preferred	0 of 20 (10 the shorter text, 10 same)	no drift
padding control, round 26: the decoy preferred	0 of 20 (20 the shorter text, 0 same)	no drift
padding control, round 29: the decoy preferred	0 of 20 (20 the shorter text, 0 same)	no drift

Table 3: The judge's controls, recomputed from their answers (`judge/controls/score.mjs`). In round 9 the longer text is the old one, so its surviving the decoy means round 9's rejection holds.

5 Examples

Each example is the first moment of its round, in the round's own order, on which both runs of the judge agreed. The tables show only the sections whose text the round changed.

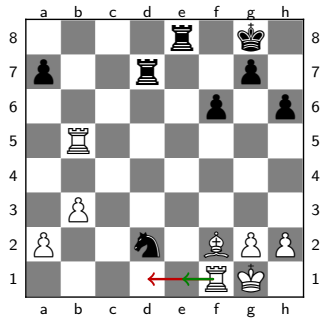


Played: 18. . . ♖xb2, Black to move; the app's verdict: Mistake.
Engine: +1.8 with the best move, +0.1 after yours.
Best line: 18. . . ♗g6 19. b4 h5 20. ♘e5 hxg4 21. ♘xg4
UCI: `b8b2`; best line `h5g6 b2b4 h6h5 f3e5 h5g4 e5g4`
FEN: `1rr3k1/5pp1/p1pbpq1p/P2p3b/Q2P2P1/2P2N1P/1P1N1P2/R3R1K1 b - - 0 18`

section	before	after
On the card	Rxb2 allows gxh5, and it wins the bishop outright. You lose a pawn in the engine's line. Bg6 avoids it. Mistake: it cost 15% of your winning chances.	Rxb2 allows gxh5, and it wins the bishop outright. You lose a pawn in the engine's line. Bg6 moves your bishop on h5 away, where it had no defender. Mistake: it cost 15% of your winning chances.
What Bg6 does instead	18...Bg6 19.b4 h5 20.Ne5 hxg4 21.Nxg4 Qh4 22.Kg2	it moves your bishop on h5 away, where it had no defender · 18...Bg6 19.b4 h5 20.Ne5 hxg4 21.Nxg4 Qh4 22.Kg2
The difference	After Rxb2, they win a pawn in the engine's line. Bg6 keeps it.	After Rxb2, they win a pawn in the engine's line. Bg6 keeps it: it moves your bishop on h5 away, where it had no defender.
Retry hint	*(not shown)*	Hint: it moves your bishop on h5 away, where it had no defender. The move starts on the lit square.
Retry answer	*(not shown)*	The move was Bg6. it moves your bishop on h5 away, where it had no defender

The judge (run 1, where the new text was Y): "Y says Bg6 rescues the attacked, undefended bishop on h5 that gxh5 wins, where X only says Bg6 'avoids it'."

Figure 1: Round 3: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

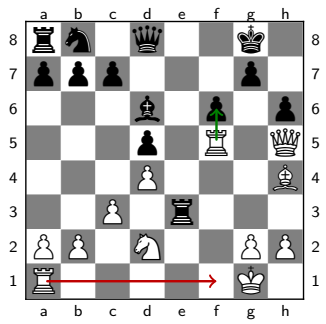


Played: 29. ♖d1, White to move; the app's verdict: Blunder.
Engine: +0.0 with the best move, −4.7 after yours.
Best line: 29. ♖e1 ♘e4 30. ♖b4 f5 31. ♖a4 ♖ee7
UCI: `f1d1`; best line `f1e1 d2e4 b5b4 f6f5 b4a4 e8e7`
FEN: `4r1k1/p2r2p1/5p1p/1R6/8/1P6/P2n1BPP/5RK1 w - - 1 29`

section	before	after
On the card	Rd1 allows Nf3+, and it attacks the pawn on h2. You lose a piece in the engine's line. Re1 attacks the rook on e8 (undefended). Blunder: it cost 35% of your winning chances.	Rd1 allows Nf3+, and if you take it with gxf3, Rxd1+ takes your rook on d1. You lose a piece in the engine's line. Re1 attacks the rook on e8 (undefended). Blunder: it cost 35% of your winning chances.
Takeaway	Before you move, ask what their best reply attacks. Nf3+ was the answer to Rd1, and it attacks the pawn on h2.	Before you move, ask what their best reply attacks. Nf3+ was the answer to Rd1, and it if you take it with gxf3, Rxd1+ takes your rook on d1.
What Rd1 allowed: Nf3+	it attacks the pawn on h2; it comes with check · 29.Rd1 Nf3+ 30.gxf3 Rxd1+ 31.Kg2 Rd2 32.Kg3	if you take it with gxf3, Rxd1+ takes your rook on d1; it attacks the pawn on h2; it comes with check · 29.Rd1 Nf3+ 30.gxf3 Rxd1+ 31.Kg2 Rd2 32.Kg3

The judge (run 1, where the new text was X): "X names the real tactic, that Nf3+ comes with check and uncovers the d7 rook so Rxd1+ wins your rook on d1, while Y only says the knight attacks h2, which does not explain the lost material."

Figure 2: Round 4: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

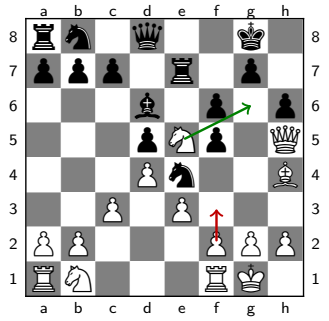


Played: 17. ♖af1, White to move; the app's verdict: Blunder.
Engine: +2.5 with the best move, 0.0 after yours.
Best line: 17. ♖xf6 ♕e8 18. ♕xd5+ ♔h8 19. ♖f7 ♘c6
UCI: `a1f1`; best line `f5f6 d8e8 h5d5 g8h8 f6f7 b8c6`
FEN: `rn1q2k1/ppp3p1/3b1p1p/3p1R1Q/3P3B/2P1r3/PP1N2PP/R5K1 w - - 0 17`

section	before	after
What Rxf6 does instead	it attacks the bishop on d6 and the pawn on h6 · 17.Rxf6 Qe8 18.Qxd5+ Kh8 19.Rf7 Nc6 20.Raf1 Ne7	it attacks the bishop on d6 and the pawn on h6; if they take it with gxf6, Qg6+ Kf8 Qxh6+ Kf7, then Qxe3 takes their rook on e3, and you are two pawns' worth of material up; if they take it with Qxf6, Bxf6 wins their queen for your rook · 17.Rxf6 Qe8 18.Qxd5+ Kh8 19.Rf7 Nc6 20.Raf1 Ne7
Retry answer	The move was Rxf6 (the green arrow). It attacks the bishop on d6 and the pawn on h6.	The move was Rxf6 (the green arrow). It attacks the bishop on d6 and the pawn on h6. If they take it with gxf6, Qg6+ Kf8 Qxh6+ Kf7, then Qxe3 takes their rook on e3, and you are two pawns' worth of material up. If they take it with Qxf6, Bxf6 wins their queen for your rook.

The judge (run 1, where the new text was X): "X shows why the rook sacrifice works (gxf6 Qg6+ Kf8 Qxh6+ then Qxe3 wins the rook back, and Qxf6 Bxf6 wins the queen), while Y leaves the player wondering why Rxf6 does not just lose a rook to gxf6."

Figure 3: Round 7: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

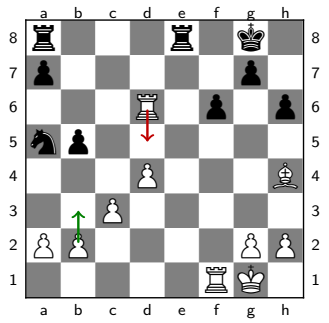


Played: 13. f3, White to move; the app's verdict: Inaccuracy.
Engine: +1.5 with the best move, +0.5 after yours.
Best line: 13. ♘g6 ♖e8 14. ♕xf5 c6 15. f3 ♘g5
UCI: `f2f3`; best line `e5g6 e7e8 h5f5 c7c6 f2f3 e4g5`
FEN: `rn1q2k1/ppp1r1p1/3b1p1p/3pNp1Q/3Pn2B/2P1P3/PP3PPP/RN3RK1 w - - 0 13`

section	before	after
Takeaway	Look for the more active move first: Ng6 attacks the rook on e7. f3 can often still be played a move later.	Look for the more active move first: Ng6 attacks the rook on e7. f3 still comes, two moves later, in the engine's line.
The difference	You win a pawn in the next 5 moves either way, so the difference is positional: Ng6 attacks the rook on e7.	You win a pawn in the next 5 moves either way, so the difference is positional: Ng6 attacks the rook on e7. The order matters: f3 comes in the engine's line too, two moves later (Ng6 Re8 Qxf5 c6 f3).

The judge (run 1, where the new text was X): "X says correctly that f3 still comes two moves later in the engine's line (Ng6 Re8 Qxf5 c6 f3), which shows the problem is move order; Y's generic 'can often still be played a move later' is vaguer and gets the count wrong."

Figure 4: Round 8: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

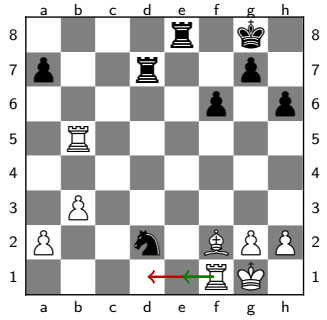


Played: 23. ♖d5, White to move; the app's verdict: Inaccuracy.
Engine: +2.1 with the best move, +1.0 after yours.
Best line: 23. b3 ♖e2 24. ♖a6 ♘b7 25. ♖e1 ♖c2
UCI: `d6d5`; best line `b2b3 e8e2 d6a6 a5b7 f1e1 e2c2`
FEN: `r3r1k1/p5p1/3R1p1p/np6/3P3B/2P5/PP4PP/5RK1 w - - 1 23`

section	before	after
What b3 does instead	if they take it with Nxb3, axb3 wins their knight for your pawn · 23.b3 Re2 24.Ra6 Nb7 25.Re1 Rc2 26.Re7 Nd8	23.b3 Re2 24.Ra6 Nb7 25.Re1 Rc2 26.Re7 Nd8
Retry answer	The move was b3 (the green arrow). If they take it with Nxb3, axb3 wins their knight for your pawn.	The move was b3 (the green arrow).

The judge (run 1, where the new text was X): "Y adds the true safety check that b3 does not hang to the a5 knight, since Nxb3 axb3 wins the knight."

Figure 5: Round 9: the judge preferred the old text in both runs. The move played is the red arrow, the engine's best move the green one.

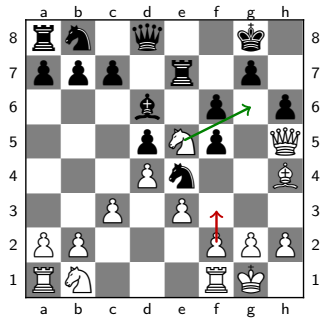


Played: 29. ♖d1, White to move; the app's verdict: Blunder.
Engine: +0.2 with the best move, −4.6 after yours.
Best line: 29. ♖e1 ♘e4 30. ♖a5 ♖ee7 31. b4 ♘xf2
UCI: `f1d1`; best line `f1e1 d2e4 b5a5 e8e7 b3b4 e4f2`
FEN: `4r1k1/p2r2p1/5p1p/1R6/8/1P6/P2n1BPP/5RK1 w - - 1 29`

section	before	after
What Re1 does instead	it attacks the rook on e8 (undefended); the rook takes the open e-file · 29.Re1 Ne4 30.Ra5 Ree7 31.b4 Nxf2 32.Rxe7 Nh3+	it attacks the rook on e8 (undefended); the rook takes the open e-file; if they take it with Rxe1+, Bxe1 takes back: an even trade · 29.Re1 Ne4 30.Ra5 Ree7 31.b4 Nxf2 32.Rxe7 Nh3+

The judge (run 1, where the new text was X): "X adds the true point that if Black trades with Rxe1+, Bxe1 recaptures for an even trade, which shows why Re1 is safe."

Figure 6: Round 10: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.

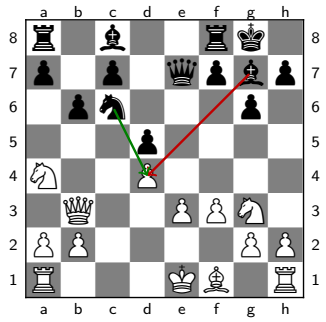


Played: 13. f3, White to move; the app's verdict: Mistake.
Engine: +1.6 with the best move, +0.3 after yours.
Best line: 13. ♘g6 ♖e8 14. ♕xf5 c6 15. f3 ♘g5
UCI: `f2f3`; best line `e5g6 e7e8 h5f5 c7c6 f2f3 e4g5`
FEN: `rn1q2k1/ppp1r1p1/3b1p1p/3pNp1Q/3Pn2B/2P1P3/PP3PPP/RN3RK1 w - - 0 13`

section	before	after
What Ng6 does instead	it attacks the rook on e7 · 13.Ng6 Re8 14.Qxf5 c6 15.f3 Ng5 16.Bf2 a5	it attacks the rook on e7; if they move the rook (Rf7), Qxf5, and you have won a pawn · 13.Ng6 Re8 14.Qxf5 c6 15.f3 Ng5 16.Bf2 a5

The judge (run 1, where the new text was X): "X adds the concrete point that once the attacked rook moves (Rf7), Qxf5 wins the f5 pawn, tying Ng6's threat to the material it gains."

Figure 7: Round 12: the judge preferred the new text in both runs. The move played is the red arrow, the engine's best move the green one.



Played: 14. . . ♗xd4, Black to move; the app's verdict: Inaccuracy.
Engine: +5.0 with the best move, +3.5 after yours.
Best line: 14. . . ♘xd4 15. ♕d3 ♗f5 16. ♕d2 ♘c2+ 17. ♔f2
UCI: `g7d4`; best line `c6d4 b3d3 c8f5 d3d2 d4c2 e1f2`
FEN: `r1b2rk1/p1p1qpbp/1pn3p1/3p4/N2P4/1Q2PPN1/PP4PP/R3KB1R b KQ - 1 14`

section	before	after
What Nxd4 does instead	it attacks the queen on b3 and the pawn on f3; it wins the pawn outright; if they move the queen (Qxd5), Qxe3+ Be2 Nc2+ Kf1 Bd4 Ne4 Bf5 Nac3 Nxa1, and you have won about 6 pawns of material · 14...Nxd4 15.Qd3 Bf5 16.Qd2 Nc2+ 17.Kf2 Nxa1 18.Nxf5	it attacks the queen on b3 and the pawn on f3; it wins the pawn outright; if they move the queen (Qxd5), you have won about 6 pawns of material by move 19 · 14...Nxd4 15.Qd3 Bf5 16.Qd2 Nc2+ 17.Kf2 Nxa1 18.Nxf5

The judge (run 1, where the new text was X): "Y shows the forcing sequence after Qxd5 (Qxe3+, Nc2+ and finally Nxa1) that wins the rook, while X only says it happens 'by move 19'."

Figure 8: Round 13: the judge preferred the old text in both runs. The move played is the red arrow, the engine's best move the green one.

6 Results

From the baseline to the kept variant (v8), the last one graded on the rounds' own documents, on the held-out games:

- faithful: 90.1 [84.2, 95.9]% to 100.0 [100.0, 100.0]%, a paired change of +9.9 [4.0, 15.9] points;
- concrete: 75.0 [65.3, 83.6]% to 82.9 [72.0, 91.4]%, a paired change of +7.9 [3.9, 12.1] points;
- named_deep: 94.1 [89.7, 97.9]% throughout, as expected of a property of the evidence;
- faithfulAll: 89.5 [83.3, 95.2]% to 99.3 [97.8, 100.0]%.

On train the same changes read 87.6% to 99.4% (faithful, +11.9 [7.9, 16.4]) and 78.0% to 84.7% (concrete, +6.8 [3.2, 10.6]). The kept variant changes the visible texts of 70 test and 87 train moments. Every kept round the judge read (all but rounds 1, 2, 11, 14 and 28, which fix errors) was preferred on both halves; round 5 was not, and rounds 9 and 13 were rejected nearly unanimously.

Round 10's sentence makes no claim the grades check. Its round was graded on the 329 moments of documents rebuilt for it, against its own base on the same documents, and the two agree on every grade (Table 4); its texts differ from the base's on 6 test and 10 train moments. So the rates above stop at v8.

Round 11 was graded the same way, on round 10's rebuilt documents against round 10's texts. It changes the texts of 35 test and 42 train moments; no grade falls, and the moments graded concrete rise from 126 to 127 on test and from 148 to 149 on train (Table 4). Of the 59 defects `defects.mjs` counts in round 10's texts, none is left (Table 6).

Round 12 was graded on documents rebuilt for it at round 11's code, with fresh capture searches and its own defense searches, against round 11's texts on the same documents. It changes the texts

of 17 test and 14 train moments, each of them graded faithful, and no grade moves (Table 4). The judge preferred the new texts 19–9 on train and 22–12 on test, with no drift toward length in the padding control dealt into the same batches. Its reasons against the new texts were mostly their length: pooling both halves, the new text won 25–5 where the sentence shows 4 moves or fewer, and 16–16 where it shows 5 or more. A round that shortens the line has seen that tally, so it fixes its cut on train.

Round 13 did, and was graded on the documents the reproduction rebuilt at the app then pinned (Section 7), against that app's texts on them, after the grader learned its two new forms and graded the base exactly as before. It changes the texts of 14 test and 8 train moments, each graded faithful, and no grade moves (Table 4). It barely shortened them: by a median of 14 characters, and 7 of the 22 grew, where the pointer is longer than the short line it replaced. The judge preferred the old texts, 0–16 on train and 4–24 on test: 1–21 where the sentence points to the engine's line, and 3–19 where it leaves out a long line. Its reasons name the moves: they show how the material falls, and a move number gives the player nothing to follow or check. The pointer was also worse than it looked: in 4 of the 11 sentences that point to the engine's line, the gain comes after the moves the lesson prints beside it, and a searched defense's line is not on the card at all. Round 12's long lines had split the judge evenly; set against the same sentence without its moves, it wanted the moves.

Round 14 was graded on the documents round 13 was, against that app's texts on them, after the grader learned to read the pieces behind each amount and the round's new phrases. Every amount the base says is the line's net, which is what faithful checks, so its misnamed pieces passed; no grade moves with the round (Table 4). One of the grader's fixes came after it first graded the round: it read "a rook for a pawn" as "a rook" where an amount ends a sentence, and failed those amounts. With the longer phrases tried first, it grades the base exactly as before, as it did after each of its other changes. What the round fixes is counted instead (Table 7): in 51 of the base's moments an amount names pieces other than the ones that changed hands, and in 10 a piece said to fall is given back later in the line. In round 14's texts neither is left and none of round 11's defects appears, and the texts change on those 58 moments (32 test, 26 train) and on no other.

Round 15 was graded on the same documents, against the texts of the app then pinned, which carry round 14, after the grader learned the round's new clauses, each checked on the board apart from the app's code, and graded the base exactly as before. It changes the texts of 4 test and 12 train moments, each graded faithful; no grade falls, and the moments graded concrete rise from 127 to 128 on test and from 149 to 152 on train (Table 4). The judge preferred the new texts 22–1 on train, with 1 same, and 8–0 on test. One moment did not win both runs: there the reply guards a pawn the better move would have attacked, but the played move attacks that pawn too and the engine answers the better move with the same reply, so the point does not tell the two moves apart.

Round 16 was graded on the same documents, against the texts of the app then pinned, which carry round 15, and on a second set built for it. The 60 games hold few endgame moments, so the round adds an endgame set: 160 more of the player's games from the routine's archive, outside the 60, that reach the endgame by the app's divider, drawn with a fixed seed and split by game, 80 train and 80 test, before any lesson was read. Its key moments in the endgame, 219 train and 184 test, were built at that app on the lab's evidence with the routine's searches. The grader learned the round's new sentences first, each checked on the board apart from the app's code, and graded the base exactly as before in both sets. The round changes the texts of 19 train and 3 test moments of the endgame set and of 2 train and 8 test moments of the 60 games, each graded faithful; no moment loses a grade, and three become concrete (Tables 4 and 5). Judging both sets together, each moment in its own set's split, the judge preferred the new texts 37–2 on train, with 3 same,

and 18–3 on test, with 1 same. Two moments lost both runs: in one the king's walk the lesson names comes in the better line too, so it does not tell the moves apart; in the other the "walk" is a king stepping out of checks and taking back a queen.

Round 17 was graded on the same documents in both sets, against the texts of the app then pinned, which carry round 16, after the grader learned to check its sentence against the stored scores apart from the app's code and graded the base exactly as before. It changes the texts of 188 train and 159 test moments across the two sets, and no grade moves (Tables 4 and 5): every one of its sentences reads true. That is more moments than a judge can read, so the judges read a sample drawn from the changed moments' ids alone, before any was read: in each set and split, 20 taken by a seeded shuffle (seed 20261015), 80 in all. On round 17's first wording the judge preferred the new texts in every comparison it decided, 72 on train, with 8 same, and 74 on test, with 6 same. Round 17b judged again the 49 sampled moments whose wording it changes, and on the final texts the judge preferred the new ones 73–0 on train, with 7 same, and 76–0 on test, with 4 same. Every same-answer is on the form that says both moves leave the mover ahead or behind, and all came from the second run's judges, who found the sentence true but no help in seeing why the better move is better. The player's five third-set moments the draw left out were judged in the same batches and are not counted: the new text won 10 of their 10 comparisons.

Round 17c was graded on round 17's documents in both sets. Before it, 30 cards gave a move one number in round 17's sentence and another in the folded note (14 on the 60 games, 16 on the endgame set); after it, none do. It changes 130 texts, and no grade moves (Tables 4 and 5). The judge sees no such note, so the round made no run, and round 17's verdict stands for its texts.

Round 18 was graded on round 17's documents, against the texts of the app then pinned, which carry round 17c, and on two endgame sets. The first two sets hold few of the checks it names, so it reads every key moment of the endgame set, 1184 rather than its 403 endgame ones, and a second endgame set drawn for it: 130 more games from the archive, outside the 60 and the first set, that reach the endgame by the first set's rule, drawn by the same seeded shuffle and split by game, 65 train and 65 test, whose 1011 key moments were built at that app after the round's plan was written. The grader learned the sentence first, checked on the board apart from the app's code, and graded the base exactly as before in every set. The round changes the texts of 10 train and 7 test moments across the three sets, and no grade moves (Tables 4 and 5): every one of its sentences reads true. The judges read all 17, each in its own set's split, and preferred the new texts 20–0 on train and 13–1 on test.

Round 19 was graded on round 18's three sets, by a grader that learned its sentence first, checked on the board apart from the app's code, and graded the base exactly as before in every set. It changes the texts of 67 train and 84 test moments, and no grade moves (Tables 4 and 5). The judges read a seeded sample of 94 of the 151 changed moments, up to 20 from each set and split (seed 20261019), and preferred the new texts 96–1 on train and 90–0 on test.

Round 20 was graded on round 19's three sets the same way, and failed its rule on one test moment, the king's attack on a guarded piece. Round 20b's plan was written after that and before any judge ran. It changes the texts of 47 train and 43 test moments; no grade falls, and concrete rises on 10 of them (Tables 4 and 5). The judges read a seeded sample of 88 of the 90 changed moments (seed 20261020) and preferred the new texts 87–3 on train and 84–2 on test.

Round 21 was graded on round 20's three sets the same way. It changes the texts of 5 train and 6 test moments, and no grade moves (Tables 4 and 5). The seeded draw (seed 20261021) takes every one of them, and the judges read all 11 and preferred the new texts 10–0 on train and 11–1 on test.

Round 22 was graded on round 21's three sets the same way, in its first wording and its last. Each changes the texts of the same 10 train and 13 test moments, and no grade moves (Tables 4 and 5). The seeded draw (seed 20261022) takes every one of them. On the first wording the judges preferred the new texts 20–0 on train, with 0 same, and 24–0 on test, with 2 same. Round 22b judged all 23 again, and on the final texts the judges preferred the new ones in every comparison, 20–0 on train and 26–0 on test.

Round 23 was graded on round 22's three sets the same way. It changes the texts of 17 train and 16 test moments, and no grade moves (Tables 4 and 5). The seeded draw (seed 20261023) takes every one of them, and the judges read all 33 and preferred the new texts 28–6 on train and 29–3 on test.

Round 24 was graded on round 23's three sets the same way. It changes the texts of 7 train and 4 test moments, and no grade moves (Tables 4 and 5). The seeded draw (seed 20261024) takes every one of them, and the judges read all 11 and preferred the new texts 8–6 on train and 6–2 on test, each moment drawing the same verdict in both runs.

Round 25 was graded on round 24's three sets the same way. It changes the texts of 7 train and 3 test moments, all in endgames, and no grade moves (Tables 4 and 5). The seeded draw (seed 20261025) takes every one of them, and the judges read all 10 and preferred the new texts 14–0 on train and 6–0 on test, in every answer of both runs: on the six moments where the point comes first and gives the headline's reason, and on the four where it follows another.

Round 26 was graded on round 25's three sets the same way. It changes the texts of 11 train and 9 test moments, and no grade moves (Tables 4 and 5). The seeded draw (seed 20261026) takes every one of them, and the judges read all 20 and preferred the new texts 22–0 on train and 18–0 on test, in every answer of both runs. The piece it names is a knight on nine of them, a bishop on seven, a rook on three and a queen on one.

Round 28 was graded on round 26's three sets, at the app then pinned. The defect it corrects stood in 3, 3 and 10 of its base's texts in the three sets, as `defects.mjs` counts it, and stands in none of its own. It changes the texts of those 16 moments and no other, 8 train and 8 test, and no grade moves (Tables 4 and 5).

Round 29 was graded on round 28's three sets, by a grader that learned its two sentences first, checked on the board apart from the app's code, and graded the base exactly as before in every set. It changes the texts of 6 train and 2 test moments, each by the point alone, and no grade moves (Tables 4 and 5); no key moment of the three sets gets the guard's pin. The seeded draw (seed 20261029) takes every one of them, and the judges read all 8 and preferred the new texts 12–0 on train and 4–0 on test, in every answer of both runs. The pinned piece is a knight on four of them, a pawn on two, a bishop on one and a rook on one; the piece that pins it is a bishop on four, a rook on three and a queen on one.

variant	split	moments	faithful	concrete	named_deep	faithfulAll	texts changed
v10	test	152	152	126	143	151	6
v10	train	177	176	148	163	176	10
v11	test	152	152	126 / 127	143	151	35
v11	train	177	176	148 / 149	163	176	42
v12	test	152	152	127	143	151	17
v12	train	177	176	149	163	176	14
v13	test	153	153	127	144	152	14
v13	train	180	179	149	166	179	8
v14	test	153	153	127	144	152	32
v14	train	180	179	149	166	179	26
v15	test	153	153	127 / 128	144	152	4
v15	train	180	179	149 / 152	166	179	12
v16	test	153	153	128 / 129	144	152	8
v16	train	180	179	152	166	179	2
v17	test	153	153	129	144	152	65
v17	train	180	179	152	166	179	72
v17b	test	153	153	129	144	152	65
v17b	train	180	179	152	166	179	72
v17c	test	153	153	129	144	152	26
v17c	train	180	179	152	166	179	34
v18	test	153	153	129	144	152	2
v18	train	180	179	152	166	179	0
v19	test	153	153	129	144	152	5
v19	train	180	179	152	166	179	9
v20b	test	153	153	129 / 130	144	152	6
v20b	train	180	179	152	166	179	6
v21	test	153	153	130	144	152	0
v21	train	180	179	152	166	179	0
v22	test	153	153	130	144	152	1
v22	train	180	179	152	166	179	4
v22b	test	153	153	130	144	152	1
v22b	train	180	179	152	166	179	4
v23	test	153	153	130	144	152	1
v23	train	180	179	152	166	179	2
v24	test	153	153	130	144	152	2
v24	train	180	179	152	166	179	0
v25	test	153	153	130	144	152	0
v25	train	180	179	152	166	179	0
v26	test	153	153	130	144	152	4
v26	train	180	179	152	166	179	3
v28	test	153	153	130	144	152	2
v28	train	180	179	152	166	179	1
v29	test	153	153	130	144	152	0
v29	train	180	179	152	166	179	0

Table 4: A round graded on rebuilt documents, beside its base on the same documents, on the 60 games: moments graded 1, with one number where the round and its base agree. v17 is round 17's first wording, v17b its last, and v17c the last with the folded note's numbers; v22 is round 22's first wording and v22b its last. Table 5 has the endgame sets.

variant	split	moments	faithful	concrete	named_deep	faithfulAll	texts changed
v16	test	184	181	162	158	180	3
v16	train	219	213	170 / 172	204	213	19
v17	test	184	181	162	158	180	94
v17	train	219	213	172	204	213	116
v17b	test	184	181	162	158	180	94
v17b	train	219	213	172	204	213	116
v17c	test	184	181	162	158	180	29
v17c	train	219	213	172	204	213	41
v18	test	572	561	481	529	560	1
v18	train	612	598	491	573	598	8
v19	test	572	561	481	529	560	43
v19	train	612	598	491	573	598	38
v20b	test	572	561	481 / 482	529	560	19
v20b	train	612	598	491 / 493	573	598	22
v21	test	572	561	482	529	560	2
v21	train	612	598	493	573	598	3
v22	test	572	561	482	529	560	7
v22	train	612	598	493	573	598	5
v22b	test	572	561	482	529	560	7
v22b	train	612	598	493	573	598	5
v23	test	572	561	482	529	560	7
v23	train	612	598	493	573	598	9
v24	test	572	561	482	529	560	1
v24	train	612	598	493	573	598	4
v25	test	572	561	482	529	560	2
v25	train	612	598	493	573	598	5
v26	test	572	561	482	529	560	5
v26	train	612	598	493	573	598	3
v28	test	572	561	482	529	560	1
v28	train	612	598	493	573	598	2
v29	test	572	561	482	529	560	2
v29	train	612	598	493	573	598	4

Table 5: Table 4 on the endgame sets. Here the endgame set: rounds 16 to 29 (rounds 18 to 29 on all its key moments, the others on its endgame moments). The second endgame set follows.

variant	split	moments	faithful	concrete	named_deep	faithfulAll	texts changed
v18	test	494	490	407	458	490	4
v18	train	517	509	420	468	509	2
v19	test	494	490	407	458	490	36
v19	train	517	509	420	468	509	20
v20b	test	494	490	407 / 408	458	490	18
v20b	train	517	509	420 / 425	468	509	19
v21	test	494	490	408	458	490	4
v21	train	517	509	425	468	509	2
v22	test	494	490	408	458	490	5
v22	train	517	509	425	468	509	1
v22b	test	494	490	408	458	490	5
v22b	train	517	509	425	468	509	1
v23	test	494	490	408	458	490	8
v23	train	517	509	425	468	509	6
v24	test	494	490	408	458	490	1
v24	train	517	509	425	468	509	3
v25	test	494	490	408	458	490	1
v25	train	517	509	425	468	509	2
v26	test	494	490	408	458	490	0
v26	train	517	509	425	468	509	5
v28	test	494	490	408	458	490	5
v28	train	517	509	425	468	509	5
v29	test	494	490	408	458	490	0
v29	train	517	509	425	468	509	2

Table 5: Table 4 on the endgame sets, continued: rounds 18 to 29 on the second endgame set, drawn for round 18.

defect	v10	v11
a later sentence of the retry reveal in lower case	29	0
a recapture called a win “outright”	15	0
a capture or a check called “a quiet improvement”	6	0
“too short or unsettled” beside a capture the card calls outright	3	0
a lost piece worth more than the amount named	6	0

Table 6: The text defects round 11 fixes, counted in each variant’s texts on every moment of round 10’s rebuilt documents (`defects.mjs`).

defect	base	v14
an amount that names pieces other than the ones that changed hands	51	0
a piece said to fall that the line gives back	10	0
the five of Table 6	0	0

Table 7: The text defects round 14 fixes, counted in its base’s texts and round 14’s on every moment of round 13’s documents (`defects.mjs`), with round 11’s five beside them.

7 Reproduction in the lab

Recomputed. The lab recomputes every variant's aggregates from the per-moment grades the records keep (`summarize.mjs`): 27 of the 27 lines of the published aggregates match exactly. Every judge round's tally is recomputed from the judge's answers and the order keys (`judge/pairwise.mjs`), and every decision follows from the rule as recorded. One recorded reason misstates its counts: round 5's reads "train 4 new vs 6 old (test 10 vs 2): the picture hurt mates of 8 to 13 moves", while the recorded answers give 6–6 on train and 14–2 on test. The new counts it gives are those of the moments round 6 later kept; the decision is the same, since a tie on train is not a majority.

Rerun. The lab rebuilt the engine evidence for the same games on the app as it now runs (`3eb8afd`, evidence review-evidence/7), redid the capture searches on it (88 kept) and the defense searches (52 kept), and graded the app's texts. The app at that commit carries every kept round. Its texts equal the recorded texts of v29, the last kept round, on 332 of the 333 moments of round 29's documents. The one that differs is the player's move and keeps the grades recorded for it; the defense searches read its natural defense at the band the app now gives the player's opponents, not his own. The rebuilt evidence is not the hill-climb's: the review pipeline changed after the rounds, so engine lines differ in detail and some key moments move (on test, 146 of 153 moments are the same moments; on train 170 of 180). Every rate lies within the recorded v8's 95% interval (Table 8). The records are read at `248fed3`.

split	grade	recorded baseline	recorded v8	pinned app, rerun
test	faithful	90.1 [84.2, 95.9]	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
test	concrete	75.0 [65.3, 83.6]	82.9 [72.0, 91.4]	85.0 [78.1, 91.1]
test	named_deep	94.1 [89.7, 97.9]	94.1 [89.7, 97.9]	94.1 [89.7, 97.9]
test	faithfulAll	89.5 [83.3, 95.2]	99.3 [97.8, 100.0]	99.3 [97.8, 100.0]
train	faithful	87.6 [83.0, 91.7]	99.4 [98.1, 100.0]	99.4 [98.2, 100.0]
train	concrete	78.0 [70.7, 84.2]	84.7 [78.9, 89.8]	84.4 [78.2, 89.9]
train	named_deep	92.1 [86.8, 97.1]	92.1 [86.8, 97.1]	92.2 [87.0, 97.1]
train	faithfulAll	87.6 [82.9, 91.8]	99.4 [98.2, 100.0]	99.4 [98.2, 100.0]

Table 8: The recorded baseline and kept variant beside the pinned app's texts on rebuilt evidence.

8 Limitations

- The grades check what can be checked: a true and concrete text can still fail to teach. That question goes to the player on the Calibration page, and to the judge, whose agreement with the player is itself only partly measured.
- The length check shows that the judge does not reward words that say nothing. It does not show that the judge can tell a point that helps from a true point that does not, at the same length; a decoy built from true but beside-the-point facts about the position would test that.
- The judge is a model reading on the player's behalf. It is a good finder of what is missing and a poorer judge of how much it matters, which is why the rounds use its pairwise preferences and its notes, not its ratings.
- Round 12 answers one defense, the model's likeliest; when the defender's second choice is nearly as likely, the lesson leaves it unanswered.

- Round 16 names a king's walk in one line without asking whether the other line makes it too; where both do, the point does not tell the moves apart. What separates such moves is often a king cut off along a file, which no reading names yet.
- Round 17's judges read a sample of the moments it changed, 80 of 347, so its verdict is the sample's. Its sentence that both moves leave the mover ahead or behind is the one the judges did not always prefer.
- Round 18 names the square a check takes from the king without asking whether another of the mover's pieces already covers it; on four of its sentences one does, so the king's guard there held nothing, and the one answer the judges gave against the round was on such a sentence.
- Round 19 names the guard a played move gives up without asking whether the better move gives it up too; on five of its 94 judged sentences nothing of the mover's guards the square after the better move either, and the one answer the judges gave against the round was on such a sentence.
- Round 20b calls the attacked piece's move an escape even where the mover can take it at once on its new square, a sacrifice rather than an escape: five of its 88 judged sentences are such, and two of the three moments the judges held against the round among them.
- Round 21 names what the kings take and the first promotion only within the plies it reads; on one of its 11 sentences that is the mover's king's capture alone, in an ending the card's closing sentence calls lost, and the one answer the judges gave against the round was on that sentence.
- Round 22 says what the better line does first even where the difference already names the reply it stops, as nine of its 23 sentences do, and the two same-answers on its first texts were on such sentences. On two of them the queen the line drives off leaves by taking the queen that drove it, a trade rather than a chase, though the judges preferred those texts in every answer on the final texts. It says nothing where no capture explains the order, as when castling comes first.
- Round 23 calls a capture a trap even where the reply's threat stood before it: had the mover passed instead of capturing, the engine's reply would make the same capture or the same mate on twelve of its 33 sentences, and there the judges preferred the new text in 19 of their 24 answers, against 38 of 42 on the others. On four the better line plays the same capture one or two of the mover's moves later, so the piece was free, only not yet; and on the round's one miss the capture leaves the game about even, a win given back rather than a loss, and both answers on it went against the round.
- Round 24 names the race even where no race decides the position. The judges' 8 answers against it fall on four moments, three of them with the pawn named four squares or more from its last rank, while each of the seven moments they preferred names a pawn three squares or fewer from it. It counts squares, not moves: it does not say how many moves each side's plan needs where no pawn queens in the lines it reads.
- Round 25 names one of the endgame readers the answers list: the opposition and the square of the pawn keep the board's narrower readings, and a square the better move covers only through a line it opens is left out.
- Round 26 reads activity alone. The answers also name restriction, the square or line the better move takes from them, which the board's readings name on few of these lessons; and a quiet best move whose pieces gain fewer than three squares from it keeps a lesson without a reason.
- Round 28 leaves the absence out only for the difference's own sentences. Round 29's point says why the reply works, and on one of the cards its judges read the difference still says the reason is not established beside it: the point separates the two moves only where the better move does not leave the pin standing too, which the round does not ask.

- Round 29's guard's pin reaches no key moment of the three sets, so its words are checked by the app's tests alone. Its gate asks that the piece guarded the square before the move, so a piece that came to the pinned square from one where it did not gets no guard sentence, though the point can still be written for it. A later round that pads the section on what the played move allowed needs filler that restates the reply alone.
- 60 games of one player. The split keeps the claims honest about that player, not about players in general.